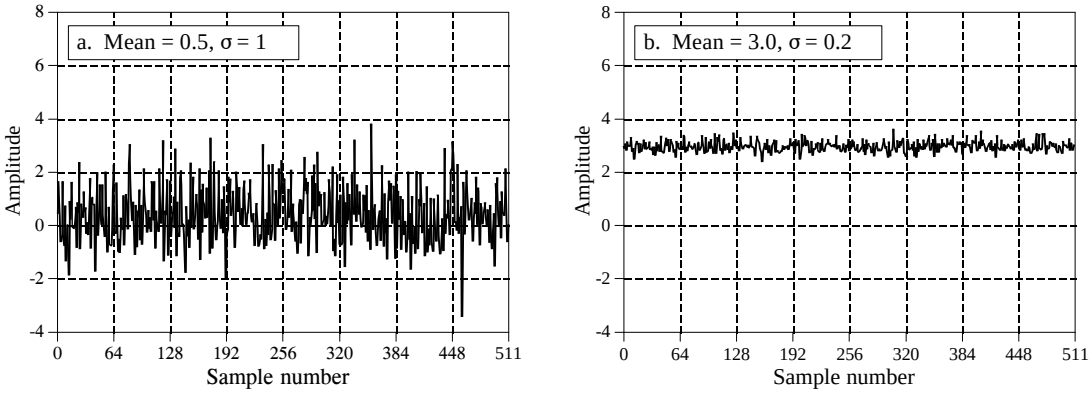

FIGURE 2-1
Examples of two digitized signals with different means and standard deviations.

keep the data organized, each sample is assigned a **sample number** or **index**. These are the numbers that appear along the horizontal axis. Two notations for assigning sample numbers are commonly used. In the first notation, the sample indexes run from 1 to N (e.g., 1 to 512). In the second notation, the sample indexes run from 0 to $N-1$ (e.g., 0 to 511). Mathematicians often use the first method (1 to N), while those in DSP commonly uses the second (0 to $N-1$). In this book, we will use the second notation. Don't dismiss this as a trivial problem. It *will* confuse you sometime during your career. Look out for it!

Mean and Standard Deviation

The **mean**, indicated by **μ** (a lower case Greek *mu*), is the statistician's jargon for the average value of a signal. It is found just as you would expect: add all of the samples together, and divide by N. It looks like this in mathematical form:

EQUATION 2-1
Calculation of a signal's mean. The signal is contained in x_0 through x_{N-1}, i is an index that runs through these values, and μ is the mean.

$$\mu = \frac{1}{N}\sum_{i=0}^{N-1} x_i$$

In words, sum the values in the signal, x_i, by letting the index, i, run from 0 to $N-1$. Then finish the calculation by dividing the sum by N. This is identical to the equation: $\mu = (x_0 + x_1 + x_2 + \cdots + x_{N-1})/N$. If you are not already familiar with Σ (upper case Greek *sigma*) being used to indicate *summation*, study these equations carefully, and compare them with the computer program in Table 2-1. Summations of this type are abundant in DSP, and you need to understand this notation fully.